

DOWNWIND AND EXPOSED

Wind-Driven Air Toxics in Commerce City, Colorado

Author & Principal Investigator

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An Independent Research Project



Why Commerce City?

Commerce City and neighboring Adams County communities sit directly downwind of Denver's refinery and fuel-terminal corridor, including the Suncor refinery. These neighborhoods are home to disproportionately impacted, largely working-class and Latinx residents who already face multiple environmental and social stressors. Colorado's Air Toxics Act (HB21-1189) created new fence-line and community monitoring for benzene, hydrogen sulfide (H₂S), and hydrogen cyanide (HCN) at nearby sites, including the A1 Commerce City monitor, making it possible to ask sharper questions about downwind exposure.

Research Question

When winds carry air from Denver's industrial corridor toward Commerce City and Adams County, do we see evidence of air toxics and fine particles in those downwind communities—and can we rule out wildfire smoke as a simple explanation during key analysis periods?

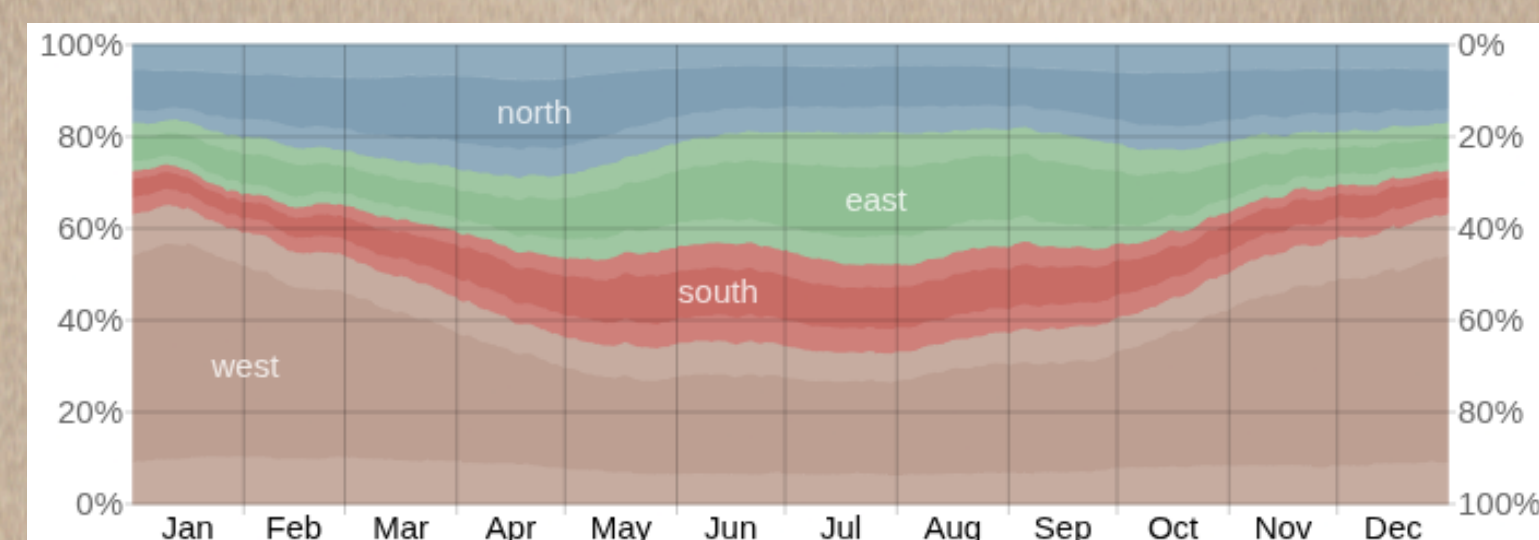


Figure 1. Fraction of hours with winds from west, south, east, and north by month. Westerly winds dominate in every season, meaning downwind communities sit beneath air arriving from the Denver-refinery corridor.

Air Toxics & Particles in Downwind Communities

- July 2023 community monitoring at the A1 Commerce City site detected benzene and H₂S at low-ppb levels in 14-day integrated samples, confirming the presence of refinery-related air toxics in the neighborhood.
- Across all three monitored regions (Commerce City, Henderson, Pueblo), July benzene ranged from 0.082 to 0.246 ppb, with Commerce City samples in the mid-range of this distribution.
- July H₂S 14-day means spanned roughly 0.15–0.71 ppb, with Commerce City values clearly above analytical detection limits.
- On 3 February 2026, the AirNow NowCast AQI map showed a moderate (yellow) pollution plume for combined ozone + particulate matter stretching from central Denver over Commerce City and Adams County (Figure 3), while upwind foothill sites remained in the “Good” range (green).

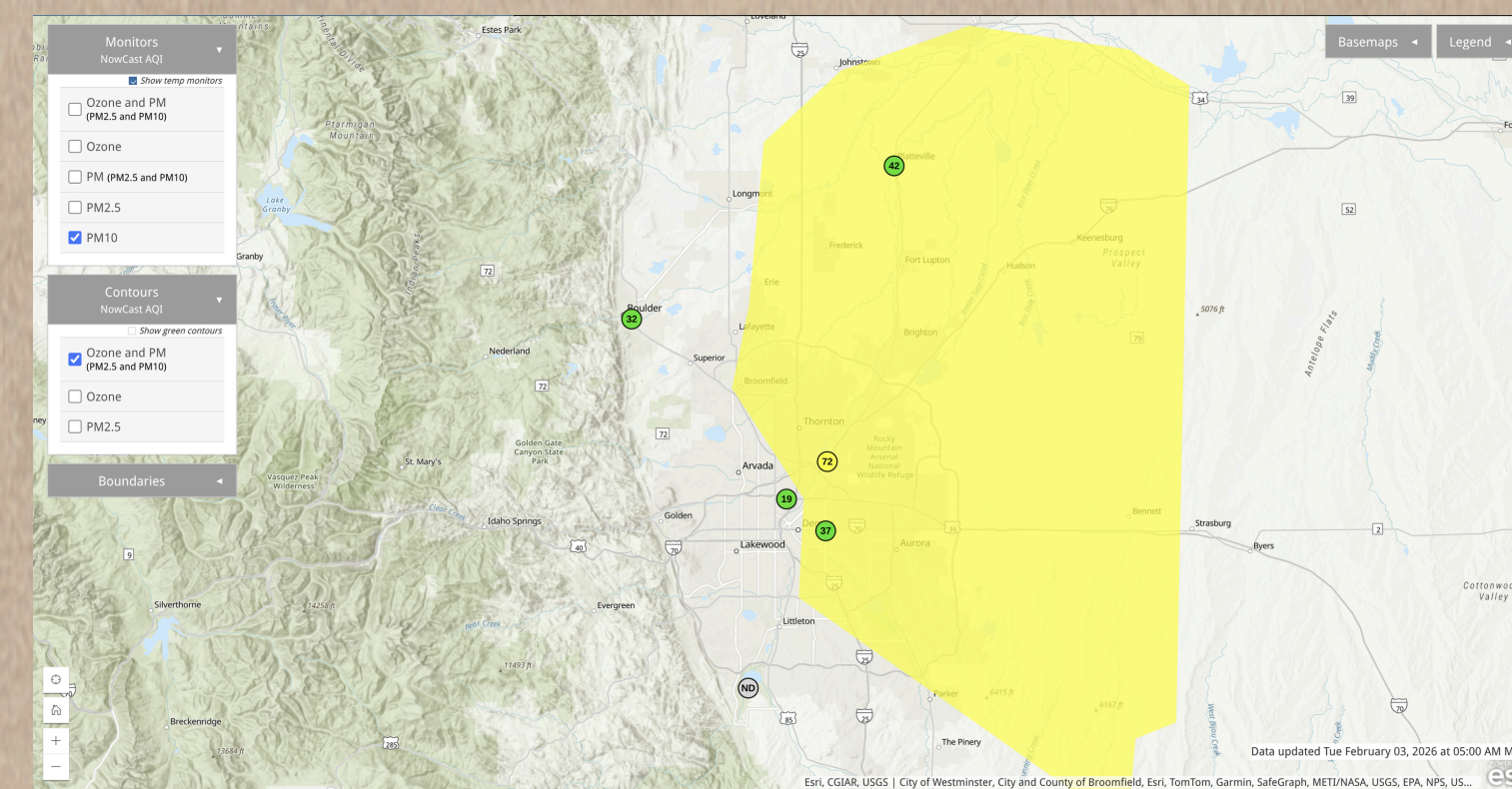


Figure 3. AirNow regional AQI map. Early-morning NowCast AQI (3 Feb 2026) shows a yellow “Moderate” pollution plume draped over the Denver–Commerce City–Adams County corridor, while foothill communities to the west remain in the “Good” range.

Where This Project Is Headed

- Expand beyond single episodes to a multi-year exposure dataset (2023–2026) integrating community air toxics, continuous sensors, HRRR reanalysis winds, and Sentinel-5P TROPOMI tropospheric NO₂ and SO₂ columns to visually map persistent pollutant plumes and define empirically affected zones.
- Because everything in Commerce City is already urban and disturbed, build an urban exposure gradient: riparian and wetland sites that are similar in urbanization but very different in how often they sit in NO₂/SO₂ plumes.
- Use spatial remote-sensing methods (NDVI, EVI, red-edge indices, NDWI, and surface reflectance time series) from Sentinel-2, Landsat 8/9, and NAIP to quantify vegetation stress, canopy change, and potential habitat degradation across that gradient.
- Deploy passive acoustic monitoring (e.g., AudioMoth units) and automated species classification models (e.g., BirdNET, CNN-based acoustic inference) to estimate avian richness and activity patterns, paired with camera trapping and field observations where feasible.
- Compare biodiversity, vegetation indices, and habitat structure metrics between high-exposure and low-exposure reference sites, and against available historical remote-sensing and survey records, to test whether chronic downwind pollutant transport corresponds with measurable ecological divergence.
- Continue developing this work independently while seeking collaboration with atmospheric chemists, urban ecologists, and remote-sensing scientists already studying refinery emissions and Front Range biodiversity dynamics.



References & Acknowledgements

Long-Term Wind Climatology

- Long-term reanalysis for Commerce City shows that westerly winds dominate throughout the year, especially from November through April.
- In most months, winds from the west account for the largest fraction of non-calm hours, with smaller contributions from south, east, and north (Figure 1).
- Average hourly wind speeds range from about 6.9 mph (August) to 9.4 mph (January), with a “windy season” from November 19–April 30 when typical speeds exceed 8.2 mph (Figure 2).
- These wind directions and speeds are sufficient to carry plumes from the refinery and urban core directly into Commerce City and adjacent neighborhoods.

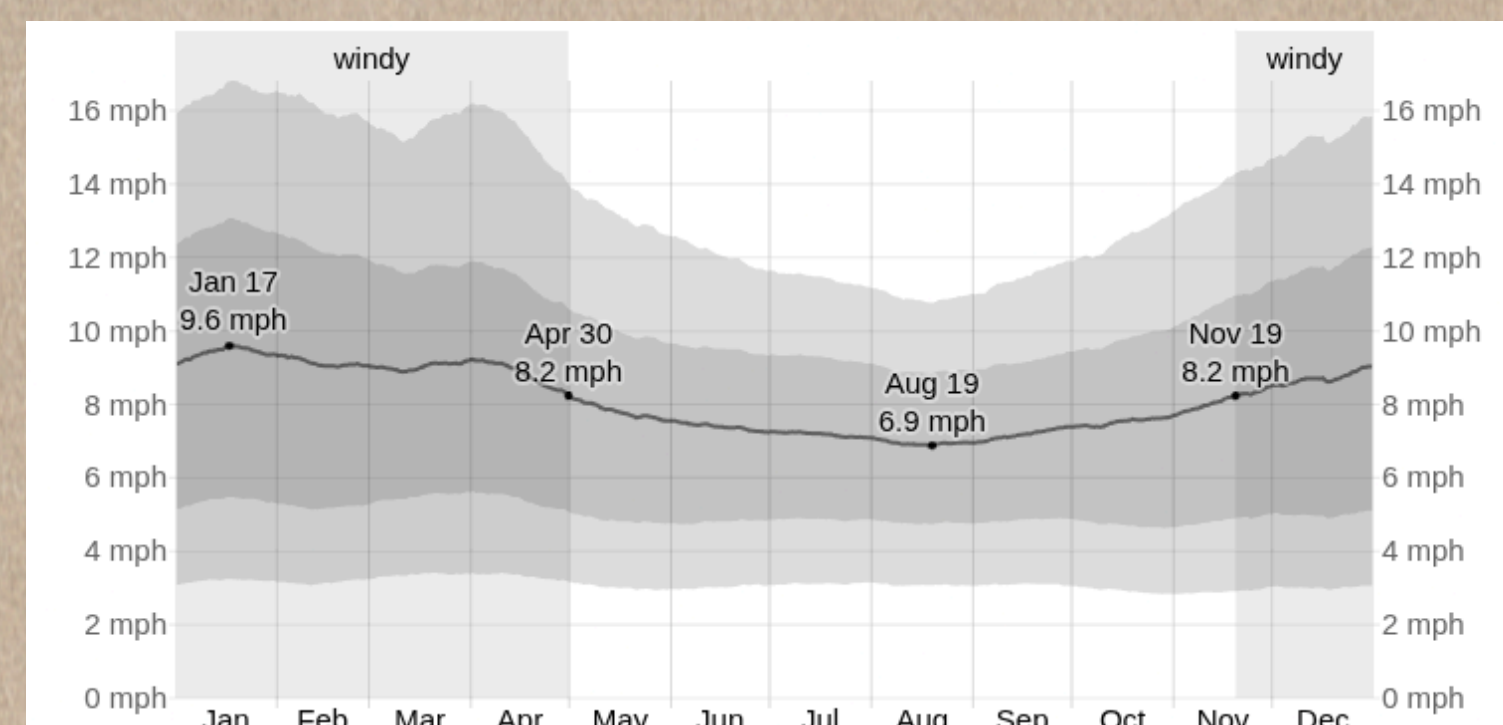


Figure 2. Seasonal cycle of hourly mean wind speed in Commerce City. Typical winds of 7–10 mph are strong enough to advect plumes several kilometers during emission events.

Ruling Out Wildfire Smoke (for Key Dates)

- Wildfire smoke can elevate PM_{2.5} across the entire Front Range. To avoid mis-attributing those events to refinery impacts, I identified “analysis windows” when smoke was not over Commerce City.
- For example, on 2–3 February 2026, the Commerce City Clarity sensor's PM_{2.5} history showed elevated but variable PM_{2.5}, while the Fire and Smoke Map panel simultaneously reported “No Nearby Fires Detected” and “No Smoke Plume Detected” for this location (Figure 4).
- Across my 2025 analysis days, a manual review of NOAA/HMS and AirNow smoke products (summarized in an internal incident log) shows multiple pollution episodes affecting Commerce City without overlapping smoke plumes, suggesting that at least some elevated PM and air toxics are driven by local industrial/urban sources rather than wildfire smoke.

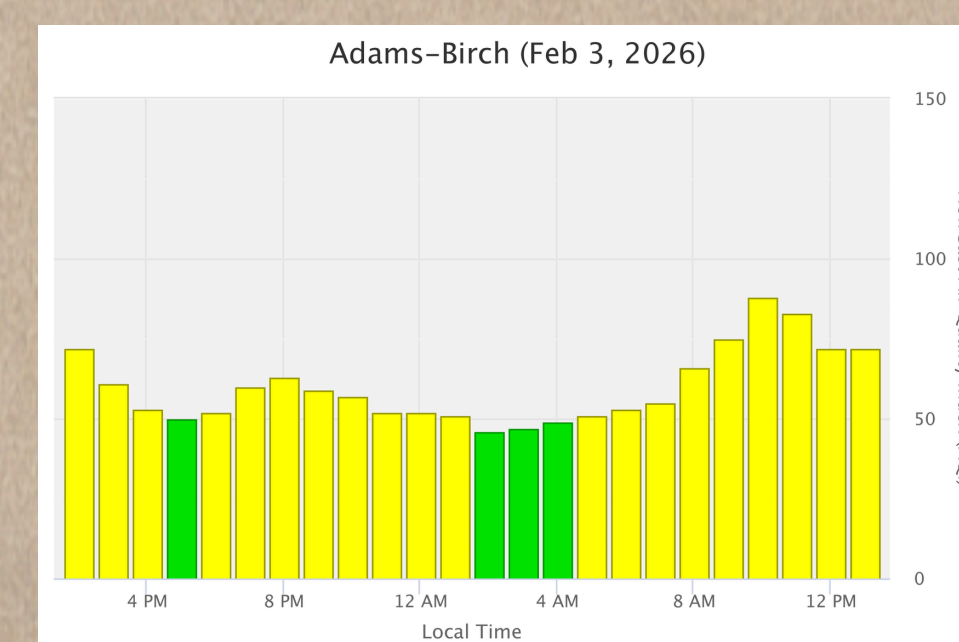


Figure 4. Commerce City PM_{2.5} and smoke status (fire.airnow.gov) Clarity sensor PM_{2.5} history for Commerce City (2–3 Feb 2026) alongside the Fire and Smoke Map legend showing “No Nearby Fires Detected” and “No Smoke Plume Detected,” indicating that this episode occurred in the absence of an overlying wildfire smoke plume.

